8.1 Practice: fractions, decimals and percentages

← Unit overview · Answers

Hi! These sets start easy and get a little harder. Do one set (or half a set) at a time. There is no clock. You can draw a strip or a ratio table (*verhoudingstabel*) whenever you like, and that is a smart move, not a cheat.

Remember: we write decimals with a comma, like 0,5 and €3,50.

Set A: One number, three outfits (Warm-up)

Fill in the missing outfits. The first one is done for you.

	Fraction (*breuk*)	Decimal (*kommagetal*)	Percentage (*procent*)
example	½	0,5	50%
1	¼		
2	¾		
3	1/10		
4		0,2	
5			30%
6	2/5		
7	1/100		
8		0,6	
9	1/8		
10	½		

Talk about it: Which row felt easiest? Which anchor (½, ¼, 1/10 …) helped you with row 6?

Set B: Which is bigger? (Warm-up → Building)

Circle the bigger number. If they're the same, write "same".

1. 0,5 or ¼
2. 30% or ¼
3. 0,7 or ¾
4. 2/5 or 45%
5. 0,35 or 1/3
6. Put in order, smallest first: 0,6 · ½ · 55%
7. Put in order, smallest first: 3/8 · 0,4 · 35%
8. Which is closest to ½? 0,45 · 48% · 5/8
9. Bram got 7 out of 10 on a quiz. Noor got 15 out of 20 on a different quiz. Who scored the higher percentage?
10. ½ or 0,50

Talk about it: For number 5, how did you think about 1/3?

Set C: Percentage of an amount (Building)

Tip: find 10% (divide by 10) or 50% (halve) first, then build.

1. 50% of €40
2. 10% of €80
3. 25% of 200
4. 10% of €35
5. 5% of €60
6. 20% of €45
7. 75% of 120
8. 1% of 600, and then 3% of 600
9. 15% of €80
10. At a school of 250 children, 30% walk to school. How many children walk?
11. 50% of €19

Talk about it: For number 9, which pieces did you build 15% from? Is there another way?

Set D: Korting and btw (Building → Stretch)

Work out the new price (*nieuwe prijs*). Remember: work out the discount, then subtract it.

1. A jacket costs €60. Today: 50% korting.
2. A game costs €40. Today: 25% korting.
3. Football boots cost €80. Today: 10% korting.
4. A bike costs €250. Today: 20% korting.
5. A sign at the shoe shop says: "Van €45 voor €36". How many percent korting is that?
6. Headphones cost €30. Today: 15% korting.
7. A winter coat costs €70. Today: 30% korting.
8. A plumber's price is €100 *excl. btw*. Btw is 21%. What do you pay *incl. btw*?
9. A bill is €200 *excl. btw*. Add 21% btw.
10. A loaf of bread costs €2,00 *excl. btw*. Food has the low btw rate of 9%. What is the price *incl. btw*?
11. A T-shirt costs €10. Today: 50% korting.

Talk about it: In number 5 you had to work backwards. How did you find the percentage?

Set E: How many percent? Up and down (Stretch)

1. 3 out of 4 children in a group like pancakes (*pannenkoeken*). What percentage is that?
2. 12 out of 50 children have a pet rabbit. What percentage is that?
3. A cinema ticket costs €20. The price goes up by 10%. What is the new price?
4. Lotte gets €5 pocket money (*zakgeld*) a week. Her parents raise it by 20%. How much does she get now?
5. A class grows from 25 to 30 children. By what percentage did it grow?
6. A bus ticket went from €3,20 to €4,00. By what percentage did the price go up?
7. A village has 4 000 people. Next year there are 5% more. How many people is that?
8. A sunflower is 40 cm tall. A week later it is 50 cm. By what percentage did it grow?
9. You get 18 out of 24 correct on a test. What percentage is that?
10. A scooter goes "van €80 voor €60". How many percent korting?
11. 1 out of 2. What percentage?

Talk about it: In numbers 5, 6 and 8, which amount was your 100%? Why that one?

Puzzle corner (optional, just for fun)

1. A shop gives 10% korting, and then at the till another 10% korting on the new price. Is that the same as 20% korting? Try it with a €100 item.
2. A price goes up by 50%, then down by 50%. Is it back where it started? Try it with €100.
3. Can something be 150%? Think of an example where that makes sense.